119TH CONGRESS **H. R. 9510**

2d Session

To amend the Federal Food, Drug, and Cosmetic Act to require an automated verification, validation, and uncertainty quantification process to clear robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation.

IN THE HOUSE OF REPRESENTATIVES

June 7, 2026

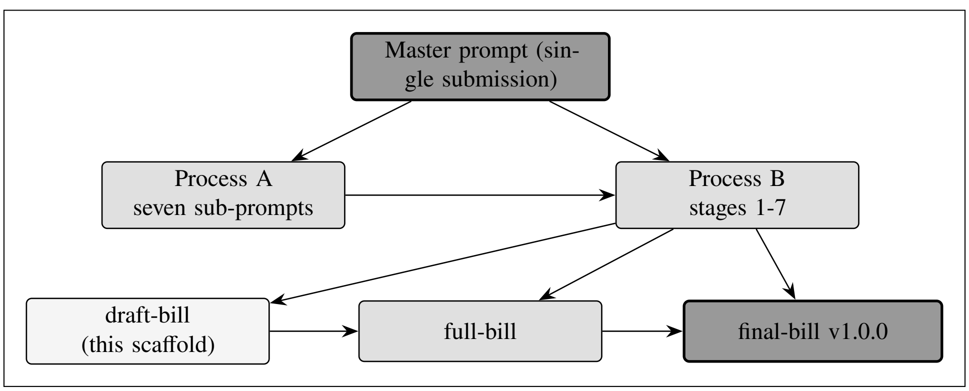
Mr. Kevin Kawchak authored the following bill with the assistance of artificial intelligence; which was referred to the Committee on Energy and Commerce.

A BILL

This legislative draft is submitted for review and consideration by the United States Congress,

---

Bill (v4.0) 10.5281/zenodo.xxxxxxxx is the gray-scale Mermaid amendment of H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026. This is the draft scaffold: it fixes the structure, the figure slots, and the twelve deliverable appendices, each carrying bracketed drafting instructions for the full bill. It was prepared with an autonomous artificial intelligence coding agent (Claude Code Opus 4.8, 1M context, Max) under human direction. It is an independent research draft, is not an enacted law, and is not legal advice.



Cover Figure. The Bill v4.0 single-prompt build: one Master prompt generates every sub-prompt (Process A), which run in sequence (Process B) to grow the bill from this draft scaffold to the full bill and the polished final bill.

SECTION 1. SHORT TITLE; TABLE OF CONTENTS.

(a) Short Title. This Act may be cited as the "Verification Before Generation in Physical AI Oncology Trials Act of 2026".

(b) Table of Contents. The table of contents for this Act version is as follows. Each entry links to the part it names.

SEC. 2. FINDINGS; THE EVIDENCE TO LAW RECORD.

(a) In General. Congress finds that an automated process must verify robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation, and that the record of the VVUQ-01 through VVUQ-04 developments establishes both the need for and the feasibility of that requirement.

[DRAFTING INSTRUCTIONS - SEC. 2. Render the fourteen findings of Bill v2.0 as a lettered narrative (a) through (g) carried by three visuals, in place of a long numbered list. Process `cancer-automated/.../VVUQ-05/final-bill/sections/s2-findings.tex` for the finding text and `.../deliverables/04-legislative-findings.md` for the authorities. Render Figure 1 (four-work evidence-to-law lineage, gray-scale Mermaid flowchart LR), Table 1 (the four works at a glance, full-width), and Figure 2 (accelerated timeline versus conventional, paired flowchart) per `auto-bill-01/04-figure-selection`. Bind findings to in-force authorities: section 515C / 21 U.S.C. 360e-4, the QMSR (part 820), ASME V&V 40, ISO/TS 15066, NASA-STD-7009A, the Medicare Advantage AI guardrail, and the State human-over-AI statutes.]

[FIGURE 1 SLOT - four-work lineage. TABLE 1 SLOT - four works at a glance. FIGURE 2 SLOT - accelerated timeline. Each is rendered in the full bill.]

SEC. 3. AMENDMENT TO THE FEDERAL FOOD, DRUG, AND COSMETIC ACT.

(a) In General. Subchapter V of the Federal Food, Drug, and Cosmetic Act (21 U.S.C. 351 et seq.) is amended by inserting after section 515C (21 U.S.C. 360e-4) a new section 515D (21 U.S.C. 360e-5) requiring verification before generation of robot-patient interaction code.

[DRAFTING INSTRUCTIONS - SEC. 3. Carry the full operative new section 515D (subsections (a) Requirement through (j) Rule of Construction), the ten conforming amendments, the clerical amendment, the rule of construction, and the effective date, so the bill stands on its own. Process `cancer-automated/.../VVUQ-05/final-bill/sections/s3-amendment.tex`. Render Table 2 (ten-gate threshold schedule, full-width), Figure 3 (gate decision rule and funnel, gray-scale Mermaid BPMN flowchart), Figure 4 (statutory layering through Title 21, clustered component), and Table 3 (ten conforming amendments crosswalk). Use the section symbol for every codified reference; single hyphens only.]

[TABLE 2 SLOT - ten-gate schedule. FIGURE 3 SLOT - decision rule and funnel. FIGURE 4 SLOT - statutory layering. TABLE 3 SLOT - conforming crosswalk. Each is rendered in the full bill.]

SEC. 4. COMPARATIVE PRINT; CHANGES IN EXISTING LAW.

(a) In General. This section shows, in comparative print, the changes the eleven conforming amendments make to existing law. Matter to be deleted is shown [strike: ...]; matter to be inserted is shown [insert: ...]; unchanged law is plain.

[DRAFTING INSTRUCTIONS - SEC. 4. Reproduce, in comparative print, the affected provisions of Title 21 (sections 201(h)/321(h), 301/331, 501/351, 505F/355g, 510(k)/360(k), 513/360c, 515/360e, 515C/360e-4, 520(o)/360j(o), 521/360k) with the strike and insert markers. Process `cancer-automated/.../VVUQ-05/final-bill/sections/s4-comparative.tex` and the crosswalk in SEC. 3. Render Table 4 (per-section comparative-print change map) and Figure 5 (the affected sections in comparative-print order, gray-scale Mermaid component flowchart). Mirror deliverable Appendix E (Ramseyer print).]

[TABLE 4 SLOT - per-section change map. FIGURE 5 SLOT - section order. The reproduced provisions and markers are rendered in the full bill.]

APPENDIX A. ONE-PAGE SUMMARY (NON-OPERATIVE).

This appendix is the one-page plain-English summary: the problem, the rule, the new section, the evidence, and the ask, on one page.

[DRAFTING INSTRUCTIONS - APPENDIX A. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 executive one-pager genre (labeled blocks: Problem, Rule, New section 515D, Evidence, The ask). Render Table 5 (one-page summary block table). Improve and update for 2026; keep it to one page. No figure required.]

APPENDIX B. SECTION-BY-SECTION ANALYSIS (NON-OPERATIVE).

This appendix explains, in plain English, every operative section, subsection, and conforming amendment of the bill.

[DRAFTING INSTRUCTIONS - APPENDIX B. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 per-section annotation genre (a gloss under each `SEC.` and each subsection (a)-(j), plus the ten conforming amendments). Mirror SEC. 2, SEC. 3, and SEC. 4. No figure required.]

APPENDIX C. PLAIN-ENGLISH POLICY MEMORANDUM (NON-OPERATIVE).

This appendix is the problem-and-solution policy memorandum: why the sequencing gap matters and why the gate is feasible and fits existing law.

[DRAFTING INSTRUCTIONS - APPENDIX C. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 policy memorandum genre (TO / FROM / RE header and a short question-and-answer body). Render Figure 8 (the verification gate as a system, gray-scale Mermaid C4 architecture) and Figure 6 (pre-introduction path, swimlane).]

APPENDIX D. LEGISLATIVE FINDINGS (NON-OPERATIVE).

This appendix is the standalone findings of fact, each tied to an in-force authority.

[DRAFTING INSTRUCTIONS - APPENDIX D. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 findings-of-fact genre (numbered findings, each with its authority). Render Table 6 (numbered findings to authority map). These findings are the long form of SEC. 2.]

APPENDIX E. RAMSEYER COMPARATIVE PRINT (NON-OPERATIVE).

This appendix shows the changes to existing law for the affected Title 21 sections, marked. Matter to be deleted is shown [strike: ...]; matter to be inserted is shown [insert: ...].

[DRAFTING INSTRUCTIONS - APPENDIX E. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 Ramseyer redline genre (`\cprstrike` and `\cprinsert`). Reproduce the eleven affected sections. This appendix is the deliverable form of SEC. 4; keep them consistent.]

APPENDIX F. CONSTITUTIONAL AUTHORITY STATEMENT (NON-OPERATIVE).

This appendix is the Constitutional Authority Statement required by House Rule XII, clause 7(c): the Commerce Clause basis on which the Federal Food, Drug, and Cosmetic Act rests.

[DRAFTING INSTRUCTIONS - APPENDIX F. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 formal-statement genre with the operative sentence in a `\billbox`. Cite Article I, section 8, clause 3. No figure required.]

APPENDIX G. PAYGO AND COST ESTIMATE NOTE (NON-OPERATIVE).

This appendix is the PAYGO-neutral note and the negative earmark declaration.

[DRAFTING INSTRUCTIONS - APPENDIX G. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 fiscal-note genre. Render Table 7 (PAYGO and cost line items) and a negative earmark declaration. State that formal CBO scoring follows introduction. No figure required.]

APPENDIX H. SPONSOR AND COSPONSOR PACKET (NON-OPERATIVE).

This appendix is the sponsor and original cosponsor packet: the sponsor role, the cosponsor form, and the referral expectation.

[DRAFTING INSTRUCTIONS - APPENDIX H. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 recruitment-packet genre (a checklist with `\cboxx` and signature lines with `\sigline`). State that H. R. 9510 is illustrative and only a sitting Member may introduce. No figure required.]

APPENDIX I. STAKEHOLDER ENGAGEMENT PLAN (NON-OPERATIVE).

This appendix is the respectful outreach plan to agencies, advocacy, industry, and standards bodies.

[DRAFTING INSTRUCTIONS - APPENDIX I. Convert `cancer-automated/.../VVUQ-05/final-bill/delivera` to LaTeX using the output-03 engagement-plan genre (a stakeholder register and phased outreach). Render Table 8 (stakeholder register and phases). Keep all outreach consistent with the not-endorsed-by framing. No figure required.]

APPENDIX J. LEGISLATIVE COUNSEL ROUTING MEMO (NON-OPERATIVE).

This appendix is the House Office of the Legislative Counsel routing memo: the legislative-form questions to settle before introduction.

[DRAFTING INSTRUCTIONS - APPENDIX J. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 routing-memo genre (an amendatory-instruction table and numbered questions). Render Table 9 (amendatory instruction table). Echo Figure 6 (pre-introduction path). Address proper amendatory form for new section 515D.]

APPENDIX K. CURRENCY AND CROSS-REFERENCE MATRIX (NON-OPERATIVE).

This appendix is the currency and cross-reference verification matrix: re-verification that every citation and amendment still lands before introduction.

[DRAFTING INSTRUCTIONS - APPENDIX K. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 audit-matrix genre (a verification table and a sign-off block). Render Table 10 (currency and cross-reference matrix) and Figure 9 (prompt and bill-version evolution, hybrid flowchart). Confirm current through Public Law 119-93 and that section 360e-5 is unused.]

APPENDIX L. TESTIMONY AND RESEARCH-INFLUENCE BRIEF (NON-OPERATIVE).

This appendix is the hearing testimony and the one place emerging bills are discussed as research influences, kept out of the operative text.

[DRAFTING INSTRUCTIONS - APPENDIX L. Convert `cancer-automated/.../VVUQ-05/final-bill/deliver` to LaTeX using the output-03 hearing-testimony genre (an opening statement, a question-and-answer block, and an influences table). Render Figure 7 (evidence to introduction handoff, gray-scale Mermaid sequence) and Table 11 (research-influence matrix). The emerging 119th Congress bills appear here only, never in SEC. 3.]

AUTHORSHIP, NOTICES, AND PROVENANCE

Prepared by CEO Kevin Kawchak (ORCID 0009-0007-5457-8667), ChemicalQDevice (kevink@chemicalqdevice.com). Adapted using Claude Code Opus 4.8 (1M context) Max; research assisted by ChatGPT and Google Gemini.

Notice: This measure was adapted from documents of the United States Government in the public domain under 17 U.S.C. 105. The reproduced statutory text is in the public domain; the authoritative version is the United States Code as published by the Office of the Law Revision Counsel. This work is not endorsed or sponsored by the FDA, HHS, the OLRC, CFR, ICH, or any member of Congress.

Disclaimer: This work is an independent research draft, is not an enacted law, and is not legal advice; it is not endorsed, sponsored, or approved by any trial sponsor, contract research organization, site, institutional review board, regulator, or medical society. All supporting numbers are simulation or illustrative results from the seed 20260525, and no patient-identifiable information is included.

DOI current bill (v4.0): 10.5281/zenodo.xxxxxxxx

DOI prior bill (v3.0): 10.5281/zenodo.xxxxxxxx

DOI repository: 10.5281/zenodo.xxxxxxxx

San Diego. June 7, 2026.

Provenance and Research Sources

References

[1] Office of the Law Revision Counsel, “Download the united states code,” 2026. Source portal for the Title 21 USLM XML reproduced in the comparative print (urldate 2026-05-31). https://uscode.house.gov/download/download.shtml.

[2] Office of the Law Revision Counsel, “United states code, title 21 - food and drugs (uslm xml),” 2026. Authoritative source of the reproduced existing-law text, current through Public Law 119-93. https://uscode.house.gov.

[3] United States Congress, “Public law 119-93 (the publication point title 21 is current through),” 2026. The public law through which the reproduced Title 21 text is current; named as the currency point, not as the law being amended. https://www.govinfo.gov/app/collection/plaw.

[4] United States Congress, “Federal food, drug, and cosmetic act, 21 u.s.c. § 301 et seq.,” 1938. The underlying Act this bill amends; the device-regulation backbone reproduced in the comparative print. https://www.law.cornell.edu/uscode/text/21/chapter-9.

[5] United States Congress, “Definitions; the term “device”, 21 u.s.c. § 321(h) (fd&c act § 201(h)),” 2021. The threshold device definition, redesignated (h)(1) by Pub. L. 116-304; the gate that decides whether robot-patient interaction code is regulated. https://www.law.cornell.edu/uscode/text/21/321.

[6] United States Congress, “Prohibited acts, 21 u.s.c. § 331 (fd&c act § 301),” 2022. The list of prohibited acts to which the Act’s penalties under 21 U.S.C. § 333 attach; the enforcement hook for the new duty. https://www.law.cornell.edu/uscode/text/21/331.

[7] United States Congress, “Adulterated drugs and devices, 21 u.s.c. § 351 (fd&c act § 501),” 2017. The device-quality backbone tying adulteration to current good manufacturing practice and 21 U.S.C. § 360j(f). https://www.law.cornell.edu/uscode/text/21/351.

[8] United States Congress, “Utilizing real world evidence, 21 u.s.c. § 355g (fd&c act § 505f),” 2016. The 21st Century Cures Act real-world-evidence program extended to recognize verification records from Physical AI oncology investigations. https://www.law.cornell.edu/uscode/text/21/355g.

[9] United States Congress, “Registration of producers; premarket notification, 21 u.s.c. § 360(k) (fd&c act § 510(k)),” 2022. The 510(k) premarket-notification pathway at § 360(k); distinct from the State-preemption section. https://www.law.cornell.edu/uscode/text/21/360.

[10] United States Congress, “Classification of devices intended for human use, 21 u.s.c. § 360c (fd&c act § 513),” 2022. The risk-based Class I, II, and III scheme and the substantial-equivalence definition; the home of graduated special controls by autonomy. https://www.law.cornell.edu/uscode/text/21/360c.

[11] United States Congress, “Premarket approval, 21 u.s.c. § 360e (fd&c act § 515),” 2022. The Class III premarket-approval pathway; the application contents amended to include the verification record. https://www.law.cornell.edu/uscode/text/21/360e.

[12] United States Congress, "Predetermined change control plans for devices, 21 u.s.c. § 360e-4 (fd&c act § 515c)," 2022. The keystone change-control authority, added by FDORA, after which the new section 515D (21 U.S.C. § 360e-5) is inserted. https://www.law.cornell.edu/uscode/text/21/360e-4.

[13] United States Congress, "General provisions; the software and clinical decision support exclusion, 21 u.s.c. § 360j(o) (fd&c act § 520(o))," 2022. The Cures Act software exclusion that robot-control software must not fall through; confirmed not to reach code that acts on a patient. https://www.law.cornell.edu/uscode/text/21/360j.

[14] United States Congress, "State and local requirements respecting devices, 21 u.s.c. § 360k (fd&c act § 521)," 1976. Device preemption; the boundary the savings clause marks for State human-over-AI requirements. Distinct from the 510(k) pathway at § 360(k). https://www.law.cornell.edu/uscode/text/21/360k.

[15] United States Congress, "Food and drug omnibus reform act of 2022 (fdora), pub. l. 117-328, div. ff, title iii," 2022. Added FD&C Act § 515C (21 U.S.C. § 360e-4), the PCP authority. https://www.congress.gov/bill/117th-congress/house-bill/2617.

[16] United States Congress, "21st century cures act, pub. l. 114-255," 2016. Source of the software and clinical decision support exclusion (21 U.S.C. § 360j(o)) and the real-world-evidence program (21 U.S.C. § 355g). https://www.congress.gov/bill/114th-congress/house-bill/34.

[17] U.S. Food and Drug Administration, "Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions," 2025. Final guidance (Dec. 4, 2024) operationalizing 21 U.S.C. § 360e-4 for AI-enabled device software; the closest existing analogue to the verification-before-change duty. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence.

[18] U.S. Food and Drug Administration, "Assessing the credibility of computational modeling and simulation in medical device submissions," 2023. Final guidance (Nov. 17, 2023) adopting the ASME V&V 40 credibility argument (model risk equals influence times consequence); the regulatory basis for the gate's credibility grading. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions.

[19] U.S. Food and Drug Administration, "Clinical decision support software (current guidance)," 2026. The controlling clinical decision support guidance; software that drives or executes a procedure cannot claim the 21 U.S.C. § 360j(o) exclusion. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software.

[20] U.S. Food and Drug Administration, "Medical devices; quality system regulation amendments (quality management system regulation), 89 fr 7496," 2024. Final rule (effective Feb. 2, 2026) revising 21 CFR part 820 to incorporate ISO 13485:2016; the live quality scaffold the adulteration tie leans on. https://www.govinfo.gov/app/details/FR-2024-02-02/2024-01709.

[21] Office of the Federal Register, "Quality management system regulation, 21 cfr part 820," 2026. The codified device quality requirement, as amended by the Quality Management System Regulation effective Feb. 2, 2026. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-820.

[22] Office of the Federal Register, "Premarket approval of medical devices, 21 cfr part 814," 2026. The premarket-approval regulation under 21 U.S.C. § 360e implementing the application-contents. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-814.

[23] Office of the Federal Register, "Medical device classification procedures, 21 cfr part 860," 2026. The classification regulation under 21 U.S.C. § 360c, including De Novo classification. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-860.

[24] Office of the Federal Register, "Electronic records; electronic signatures, 21 cfr part 11," 2026. The record-integrity basis for the hash-chained, tamper-evident audit trail required by section 515D. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11.

[25] Office for Civil Rights, HHS, "Hipaa security rule, 45 cfr part 164, subpart c," 2026. The audit-control (§ 164.312(b)) and integrity (§ 164.312(c)) requirements supporting the audit trail. https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-164.

[26] Office for Civil Rights, HHS, "Nondiscrimination in health programs and activities (section 1557 final rule), 89 fr 37522," 2024. Final rule (effective July 5, 2024) that promulgated 45 CFR § 92.210; the statutory authority is 42 U.S.C. § 18116. https://www.govinfo.gov/app/details/FR-2024-05-06/2024-08711.

[27] Office for Civil Rights, HHS, "Nondiscrimination in the use of patient care decision support tools, 45 cfr § 92.210," 2024. The in-force anchor for the bias-minimization and accessibility duties (identify-and-mitigate compliance date May 1, 2025). https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-92/subpart-B/section-92.210.

[28] United States Congress, "Section 1557 of the patient protection and affordable care act, 42 u.s.c. § 18116," 2010. The statutory nondiscrimination authority behind the 45 CFR § 92.210 rule. https://www.law.cornell.edu/uscode/text/42/18116.

[29] Centers for Medicare and Medicaid Services, "Medicare advantage and part d final rule; artificial intelligence and algorithm guardrail, 42 cfr § 422.101(c)," 2023. 88 FR 22120 (applicable for coverage beginning Jan. 1, 2024); the strongest federal precedent for mandatory human-over-AI oversight in coverage decisions. https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-422/subpart-C/section-422.101.

[30] United States Congress, "Medicare; exclusions from coverage (reasonable and necessary), 42 u.s.c. § 1395y(a)(1)(a)," 2026. A device cleared under this Act must still be reasonable and necessary for coverage; the verification record supports that determination. https://www.law.cornell.edu/uscode/text/42/1395y.

[31] American Medical Association, "Cpt appendix s: Artificial intelligence taxonomy (assistive, augmentative, autonomous)," 2026. The graduated-autonomy spine mapped to special controls; robot-control code sits at the autonomous tier requiring the strictest ex ante gate. https://www.ama-assn.org/practice-management/cpt/cpt-appendix-s-ai-taxonomy-medical-services-procedures.

[32] Assistant Secretary for Technology Policy, ONC, "Health data, technology, and interoperability: Certification program updates, algorithm transparency, and information sharing (hti-1), 89 fr 1192," 2024. The decision-support intervention transparency template (45 CFR § 170.315(b)(11)); cited as algorithm-transparency context, not as operative authority. https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-C/section-170.315.

[33] United States Congress, "Title vi of the civil rights act of 1964, 42 u.s.c. § 2000d," 1964. Nondiscrimination basis (Title 42, not Title 26) for the bias-minimization duty. https://www.law.cornell.edu/uscode/text/42/2000d.

[34] United States Congress, "Americans with disabilities act of 1990, 42 u.s.c. § 12101 et seq.," 1990. Accessibility and accommodation basis (Title 42) for the nondiscrimination duty. https://www.law.cornell.edu/uscode/text/42/12101.

[35] United States Congress, "Section 504 of the rehabilitation act of 1973, 29 u.s.c. § 794," 1973. Disability nondiscrimination basis (Title 29) complementing the accessibility duty. https://www.law.cornell.edu/uscode/text/29/794.

[36] ASME, "Asme v&v 40-2018: Assessing credibility of computational modeling through verification and validation: Application to medical devices," 2018. FDA-recognized consensus standard; the credibility basis (model risk equals influence times consequence) for the verification, validation, and uncertainty quantification gate. https://www.asme.org/codes-standards/find-codes-standards/assessing-credibility-of-computational-modeling-through-verification-and-validation-application-to-medical-devices.

[37] ASME, "Asme vvuq 70: Verification, validation, and uncertainty quantification for machine learning (in development)," 2026. The machine-learning credibility standard in development; cited as direction only. There is no ASME V&V 90. https://www.asme.org/conferences-events/events/vvuq-70-verification-validation-and-uncertainty-quantification-of-machine-learning.

[38] National Aeronautics and Space Administration, "Nasa-std-7009a: Standard for models and simulations," 2016. Treats uncertainty quantification as a first-class credibility factor (epistemic and aleatory); the basis for the coefficient-of-variation dispersion bounds. https://standards.nasa.gov/standard/nasa/nasa-std-7009.

[39] IEEE, "Ieee std 7009-2024: Standard for fail-safe design of autonomous and semi-autonomous systems," 2024. DOI: 10.1109/IEEESTD.2024.10582898. Fail-safe behavior and hand-back-to-human escalation underpinning the ESCALATE rule. https://doi.org/10.1109/IEEESTD.2024.10582898.

[40] International Electrotechnical Commission, "Iec 80601-2-77:2019: Requirements for the basic safety and essential performance of robotically assisted surgical equipment," 2019. The dedicated surgical-robot safety standard defining the essential performance the gate protects. https://www.iso.org/standard/68473.html.

[41] International Organization for Standardization, "Iso 14971:2019: Medical devices - application of risk management to medical devices," 2019. The master risk-management process; the gate is a risk-control measure and the hard catastrophe predicate a risk acceptance limit under it. https://www.iso.org/standard/72704.html.

[42] International Organization for Standardization, "Iso/ts 15066:2016: Robots and robotic devices - collaborative robots," 2016. Power-and-force-limiting and biomechanical thresholds for robot-patient contact; binds the clearance gates. https://www.iso.org/standard/62996.html.

[43] International Electrotechnical Commission, "Iec 62304:2006/amd 1:2015: Medical device software - software life cycle processes," 2015. Software safety classification (class A, B, C); its sole normative reference is ISO 14971. https://webstore.iec.ch/publication/22794.

[44] Underwriters Laboratories, "Ansi/ul 4600: Standard for safety for the evaluation of autonomous products," 2023. The claim-based safety case (claims, argument, evidence) and safe-state default for the autonomy gates. https://users.ece.cmu.edu/~koopman/ul4600/index.html.

[45] International Organization for Standardization, "Iso 13482:2014: Robots and robotic devices - safety requirements for personal care robots," 2014. Stability, collision avoidance, and fail-safe requirements for a close-contact humanoid. https://www.iso.org/standard/53820.html.

[46] International Organization for Standardization, "Iso 10218-1 and -2: Robotics - safety requirements for industrial robots," 2025. Speed-and-separation monitoring and safety-rated stops; the 2025 revision integrates the collaborative-robot limits of ISO/TS 15066. https://www.iso.org/standard/73933.html.

[47] International Electrotechnical Commission, "Iec 60601-1: Medical electrical equipment - part 1: General requirements for basic safety and essential performance," 2020. The single-fault safe-state requirement underlying the fault and emergency-stop gate. https://webstore.iec.ch/publication/67497.

[48] International Organization for Standardization, "Iso 13849-1:2023: Safety of machinery - safety-related parts of control systems," 2023. The performance level of the fault and emergency-stop control path. https://www.iso.org/standard/73481.html.

[49] International Organization for Standardization, "Iso 9283:1998: Manipulating industrial robots - performance criteria and related test methods," 1998. Pose accuracy and repeatability for the servo, grasp, and handover gates. https://www.iso.org/standard/22244.html.

[50] National Institute of Standards and Technology, "Nist ai risk management framework 1.0 (nist ai 100-1)," 2023. DOI: 10.6028/NIST.AI.100-1. Risk-management framing (map, measure, manage, govern) as direction, not operative authority. https://doi.org/10.6028/NIST.AI.100-1.

[51] Office of the Federal Register, "Investigational new drug application, 21 cfr part 312," 2026. The IND framework the verification gate attaches to and that the rule of construction preserves. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-312.

[52] Office of the Federal Register, "Protection of human subjects, 21 cfr part 50," 2026. Human-subject protection and informed consent the bill preserves unchanged. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-50.

[53] Office of the Federal Register, "Institutional review boards, 21 cfr part 56," 2026. The IRB review framework the bill preserves unchanged. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-56.

[54] International Council for Harmonisation, "Ich e6(r3): Good clinical practice," 2025. The harmonized good-clinical-practice basis for the recorded human review and audit requirements. https://www.ich.org/page/efficacy-guidelines.

[55] K. Kawchak, "Vvuq physical ai oncology trial bill (h.r. 9510, draft 1.0)," May 2026. DOI: 10.5281/zenodo.20454870. The prior bill this amendment recasts; source at papers/VVUQ-03/final-paper. https://doi.org/10.5281/zenodo.20454870.

[56] K. Kawchak, "Vvuq-01: Automated verification, validation, and uncertainty quantification of generated code," 2026. The first study: a complete VVUQ pipeline holding the verification fraction at

1.0; 51 of 51 supporting tests passed. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-01.

[57] K. Kawchak, "Vvuq-02: A ten-gate assurance suite for robot-patient interaction code," 2026. The second study: control software for a surgical humanoid cleared a ten-gate suite bound to fourteen external standards and two clinical baselines, reproducibly from a fixed seed. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-02.

[58] K. Kawchak, "National platform for physical ai oncology trials: Usl and psl readiness standards," 2026. DOI: 10.5281/zenodo.19244918. The Unification Standard Level and Physical AI Standard Level readiness standards the gates rely on. https://doi.org/10.5281/zenodo.19244918.

[59] K. Kawchak, "cancer-automated: Automated physical ai oncology trial daily deliverables," 2026. DOI: 10.5281/zenodo.20372501. The repository holding the VVUQ-01 through VVUQ-04 records and this full-bill. https://github.com/kevinkawchak/cancer-automated.

[60] K. Kawchak, "physical-ai-oncology-trials: National platform," 2026. The sibling repository with the USL and PSL standards and the trial simulations; read only as a source, not edited. https://github.com/kevinkawchak/physical-ai-oncology-trials.

[61] K. Kawchak, "Vvuq-04 instruct-bill: U.s. medical ai bill and law summaries (through may 31, 2026)," 2026. The legal crosswalk and bill-style instruction set (v2.0.0) whose ten markdown files and five .bib files ground this bill. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/instruct-bill.

[62] K. Kawchak, "Vvuq-04 template-bill: Title 21 device-provisions excerpt (latex)," 2026. The faithful Title 21 reproduction (v2.1.0) whose eleven section files form the comparative-print base. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/template-bill.

[63] K. Kawchak, "Vvuq-04 draft-bill: Fd&c act amendment scaffold with drafting instructions (latex)," 2026. The amendment scaffold (v2.2.0) whose bracketed drafting instructions this full-bill executes. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-04/draft-bill.

[64] K. Kawchak, "Vvuq-03 final-paper: Vvuq physical ai oncology trial bill source," 2026. The prior-bill LaTeX source whose operative text, definitions, and crosswalk this amendment converts into FD&C Act amendatory form. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-03/final-paper.

[65] Anthropic, "Claude code opus 4.8 (1m context) max," 2026. The coding agent that authored the v2.0.0 to v2.3.0 pipeline autonomously across sequential commits; named for development transparency. https://claude.ai/code.

[66] California State Legislature, "Sb 1120, physicians make decisions act (chapter 879, statutes of 2024)," 2024. Effective Jan. 1, 2025; bars AI in utilization review from replacing a licensed physician's medical-necessity determination. Cited as a finding of existing human-over-AI practice. https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240SB1120.

[67] Illinois General Assembly, "Hb 1806, wellness and oversight for psychological resources act (public act 104-0054)," 2025. Effective Aug. 4, 2025; bars AI from independent therapeutic decisions without licensed-professional review. Cited as a finding of existing practice. https://legiscan.com/IL/bill/HB1806/2025.

[68] Texas Legislature, "Hb 149, texas responsible artificial intelligence governance act (traiga)," 2025. Effective Jan. 1, 2026; comprehensive AI governance with a regulatory sandbox and advisory council. A finding of existing practice. https://legiscan.com/TX/bill/HB149/2025.

[69] Maryland General Assembly, "Hb 820, utilization review - use of artificial intelligence (chapter 747)," 2025. Effective Oct. 1, 2025; AI in utilization review must use individual clinical information and may not autonomously deny care. Cited as a finding of existing practice. https://mgaleg.maryland.gov/mgawebsite/Legislation/Details/hb0820.

[70] New York State Legislature, "A9149, artificial intelligence in utilization review," 2024. Would require submission of utilization-review algorithms for bias certification; the algorithm-certification model. Cited as a finding of existing practice. https://www.nysenate.gov/legislation/bills/2023/A9149.

[71] Executive Office of the President, "Executive order 14179, removing barriers to american leadership in artificial intelligence," 2025. Research influence only (90 FR 8741); deregulatory posture cited in the findings narrative and Appendix A, never in operative text. https://www.govinfo.gov/app/details/DCPD-202500170.

[72] Executive Office of the President, "Executive order 14365, ensuring a national policy framework for artificial intelligence," 2025. Research influence only (signed Dec. 11, 2025); directs review of conflicting State AI laws. Cited only in Appendix A. https://public-inspection.federalregister.gov/2025-23092.pdf?1765815309.

[73] Office of Management and Budget, "Omb memorandum m-25-21, accelerating federal use of ai through innovation, governance, and public trust," 2025. Research influence only (Apr. 3, 2025); defines high-impact AI and pre-deployment testing for federal use. Cited only in Appendix A. https://www.whitehouse.gov/omb/information-for-agencies/memoranda/.

[74] U.S. Department of Health and Human Services, "Hhs artificial intelligence strategy," 2025. Research influence only (released Dec. 4, 2025); governance and research-reproducibility pillars cited in the findings narrative and Appendix A. https://www.hhs.gov/programs/topic-sites/ai/index.html.

[75] United States Congress, "H.r. 238, healthy technology act of 2025," 2025. Research influence only; 119th Congress bill on AI prescribing authority, referred to Energy and Commerce. Not operative authority. https://www.congress.gov/bill/119th-congress/house-bill/238.

[76] United States Congress, "S. 1399, health tech investment act," 2025. Research influence only; 119th Congress bill on a Medicare payment class for algorithm-based services, referred to Finance. Not operative. https://www.congress.gov/bill/119th-congress/senate-bill/1399.

[77] United States Congress, "H.r. 6361, ban ai denials in medicare act," 2025. Research influence only; 119th Congress bill barring AI-driven prior authorization in traditional Medicare. Not operative authority. https://www.congress.gov/bill/119th-congress/house-bill/6361.

[78] United States Congress, "H.r. 5045, health ai act," 2025. Research influence only; 119th Congress bill creating an HHS grant program on generative AI in health care, referred to Energy and Commerce. Not operative authority. https://www.congress.gov/bill/119th-congress/house-bill/5045.

[79] United States Congress, "H.res. 694, expressing the sense of the house on the cms wiser artificial intelligence pilot," 2025. Research influence only; a simple resolution urging CMS to halt AI in Medicare coverage decisions. Cited by number and subject; not operative authority. https://www.congress.gov/bill/119th-congress/house-resolution/694.

[80] K. Kawchak, "Verification before generation in physical ai oncology trials act of 2026 (h.r. 9510; fd&c act amendment, bill v2.0)," June 2026. DOI: 10.5281/zenodo.20485580. Bill v2.0, the finished FD&C Act amendment this visual bill reorganizes; source at papers/VVUQ-04/final-bill. https://doi.org/10.5281/zenodo.20485580.

[81] K. Kawchak, "Verification before generation in physical ai oncology trials act of 2026 (h.r. 9510; fd&c act amendment, bill v3.0, the visual amendment)," June 2026. DOI: 10.5281/zenodo.20535429. Bill v3.0, this finished visual amendment; source at papers/VVUQ-05/full-bill. https://doi.org/10.5281/zenodo.20535429.

[82] K. Kawchak, "Two stage vvuq oncology clinical trial verification automation priority over existing generated code (vvuq-01)," 2026. DOI: 10.5281/zenodo.20372501. The method paper and pipeline; 51 of 51 tests, a 1 ACCEPT, 5 BLOCK, 1 ESCALATE gate surface, and the PDAC pilot reference. https://doi.org/10.5281/zenodo.20372501.

[83] K. Kawchak, "Mobile pancreatic cancer unitree h2 surgical humanoid with priority vvuq (vvuq-02)," 2026. DOI: 10.5281/zenodo.20421754. The ten-gate humanoid assurance suite; 172 of 172 tests and a 32 of 32 sweep at composite mean 93.56 from seed 20260525. https://doi.org/10.5281/zenodo.20421754.

[84] K. Kawchak, "Vvuq-05 update-bill figures-bill: verbatim visual catalog of the papers/ corpus," 2026. The five-file catalog (01 root, 02 VVUQ-01, 03 VVUQ-02, 04 VVUQ-03, 05 VVUQ-04) reproducing every ASCII diagram, timeline, repository tree, and table; the visual source for the figure and table slots. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-05/update-bill/figures-bill.

[85] K. Kawchak, "Vvuq-05 update-bill next-steps: pre-submission steps for a u.s. house bill (2026)," 2026. Two multi-model outputs identifying the deliverables and process a draft House bill needs before introduction; the source for the deliverables appendix. https://github.com/kevinkawchak/cancer-automated/tree/main/papers/VVUQ-05/update-bill/next-steps.

[86] U.S. House of Representatives, "Rules of the house of representatives, 119th congress (rule xii, clause 7)," 2025. Receipt and referral of measures, sponsorship and cosponsorship, and the constitutional authority statement requirement at clause 7(c)(1). https://www.congress.gov/bill/119th-congress/house-resolution/5/text.

[87] Congressional Research Service, "Introducing a house bill or resolution (crs report r44001)," 2025. The mechanics of introduction: the bill is not numbered until introduced, the Clerk assigns the number, and only a Member may introduce a House bill. https://www.everycrsreport.com/reports/R44001.html.

[88] Congressional Research Service, "Sponsorship and cosponsorship of house bills (crs report rs22477)," 2024. One sponsor and an unlimited number of cosponsors; the basis for the sponsor and cosponsor packet deliverable. https://crsreports.congress.gov/product/details?prodcode=RS22477.

[89] Congressional Research Service, "Constitutional authority statements: A quick guide (crs report if12335)," 2023. Required by House Rule XII, clause 7(c); for an FD&C Act amendment the cited power is the Commerce Clause (Art. I, sec. 8, cl. 3). https://crsreports.congress.gov/product/details?prodcode=IF12335.

[90] Office of the Legislative Counsel, U.S. House, "Understanding the legislative process; house legislative counsel drafting guide," 2026. Nonpartisan, confidential drafting services; positive-law status affects whether a bill amends the U.S. Code provision or the underlying statute. https://legcounsel.house.gov/.

[91] Office of the Clerk, U.S. House, "ehopper quick guide (electronic introduction of measures)," 2026. The House primary electronic introduction mechanism; submissions accepted near convening and adjournment; generates and submits the constitutional authority statement. https://www.house.gov/.

[92] Office of the Parliamentarian, U.S. House, "How our laws are made," 2026. The reference on the legislative process: introduction, referral, committee, and the bill versus resolution distinction. https://www.congress.gov/help/learn-about-the-legislative-process/how-our-laws-are-made.

[93] Congressional Budget Office, "Cost estimates and the statutory pay-as-you-go act of 2010," 2026. The basis for the PAYGO and cost-estimate deliverable; informal CBO consultation can precede a formal score. https://www.cbo.gov/system/files/2026-06/hr8872.pdf.

[94] National Institute of Standards and Technology, "Nist ai 600-1: Artificial intelligence risk management framework: Generative ai profile," 2024. DOI: 10.6028/NIST.AI.600-1. Generative-AI risk and transparency framing; direction for the explainability standard, not operative authority. https://doi.org/10.6028/NIST.AI.600-1.

[95] P. J. Phillips, C. A. Hahn, P. C. Fontana, A. N. Yates, K. Greene, D. A. Broniatowski, and M. A. Przybocki, "Four principles of explainable artificial intelligence (nistir 8312)," 2021. DOI: 10.6028/NIST.IR.8312. The explanation, meaningful, accuracy, and knowledge-limits principles the four-work record is mapped to in Appendix C. https://doi.org/10.6028/NIST.IR.8312.